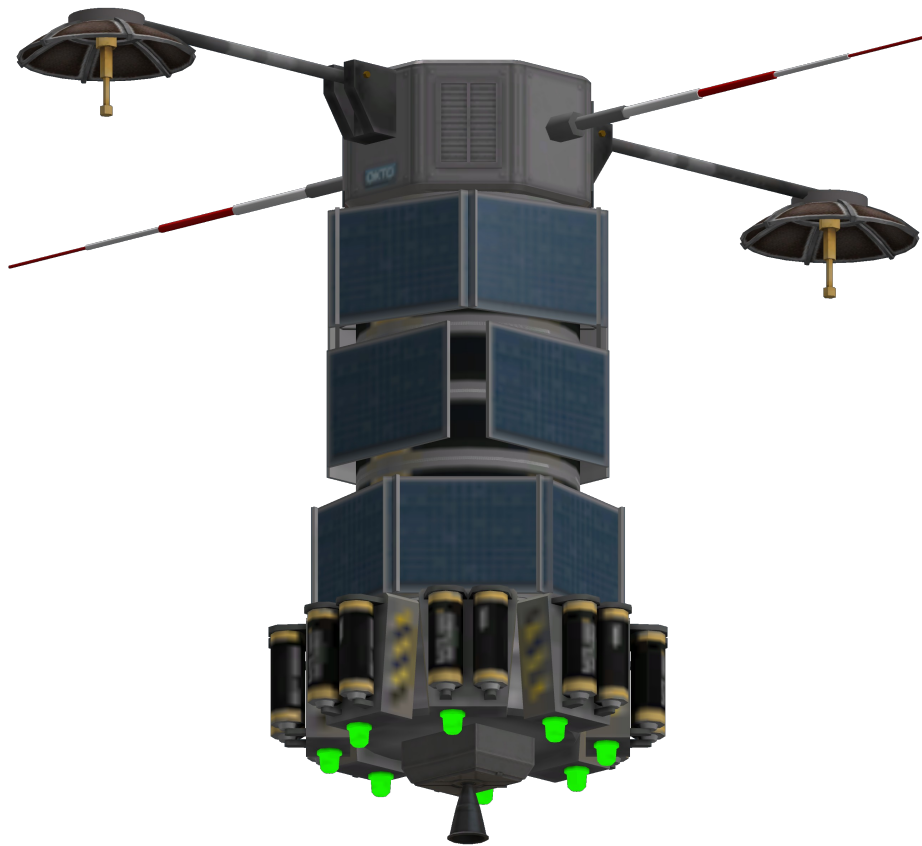


# Model 1122 communications satellite

Specifications and assembly instructions



## Description

The model 1122 satellite is an HG-5 based relay satellite that takes advantage of the wide angle of the HG-5 antenna to cover a large footprint on planetary surfaces. The presence of a second antenna allows the satellite to relay to off-world targets, to act as a close range (Mun to Kerbin) relay or as part of a larger communications network.

## Variants

The satellite variants can be identified by the following suffixes

Model 1122[Y]

Where Y defines the attachment point and can be

| Suffix      | Type                | Notes    |
|-------------|---------------------|----------|
| (no suffix) | No attachment point |          |
| K           | KAS JS-1 socket     | KAS only |
| C           | Clamp-o-Tron jr     |          |

## Uses

TSC satellite

Mun-Kerbin relay

General relay

## Costs and tech

### Costs

|            |      |       |
|------------|------|-------|
| Base model | 6430 | Funds |
| Model K    | 7180 | Funds |
| Model C    | 7230 | Funds |

### Tech requirements

| Technology         | Use                                                  | Notes |
|--------------------|------------------------------------------------------|-------|
| Electrics          | Probodobodyne OKTO<br>Z-200 Battery<br>OX-STAT Panel |       |
| Basic science      | Z-100 Battery<br>HG-5 Antenna                        |       |
| Engineering 101    | Communotron 16                                       |       |
| Propulsion systems | LV-1 Ant engine                                      |       |

|                       |                                  |              |
|-----------------------|----------------------------------|--------------|
| Miniaturization       | OSCAR-B tank<br>Clamp-O-Tron jr. | Model C only |
| Advanced construction | JS-1 socket                      | Model K only |

## Specifications

### General specifications

| Parameter                           | Value | Units     | Notes                                        |
|-------------------------------------|-------|-----------|----------------------------------------------|
| Electrical storage                  | 2010  | EC        |                                              |
| Electrical storage                  | 19    | %         | kerbostationary orbit                        |
| margin                              | 384   | EC        | kerbostationary orbit                        |
| Max orbital alt<br>(circular)       | 4725  | Km        | limited by battery storage                   |
| Min orbital alt<br>(circular)       | 575   | Km        | limited by battery charger                   |
| Power consumption<br>(absolute max) | 1.41  | EC/sec    |                                              |
| Power consumption<br>(deployment)   | 0.28  | EC/sec    | Omnis on, HG5 off, OKTO on, wheels idle      |
| Power consumption<br>(mission)      | 1.36  | EC/sec    | OKTO hibernating, wheels off, radios full on |
| Power generation<br>(max)           | 2.1   | EC/sec    | Assumes 0.4 EC for 100% exposure             |
|                                     | 5.24  | %Exposure |                                              |
| Power generation<br>(min)           | 1.94  | EC/sec    | assumes panel ring oriented towards sun      |
|                                     | 4.86  | %Exposure | assumes panel ring oriented towards sun      |
| Power generation<br>(avg)           | 2.03  | EC/sec    | assumes panel ring oriented towards sun      |
|                                     | 5.87  | %Exposure | assumes panel ring oriented towards sun      |
| Thrust                              | 2     | kN        |                                              |
| dV budget (vacuum)                  | 1147  | m/s       | 1122C                                        |
|                                     | 1173  | m/s       | 1122K                                        |
|                                     | 1191  | m/s       | 1122                                         |
| Mass (wet)                          | 695   | Kg        | 1122C                                        |
|                                     | 683   | Kg        | 1122K                                        |
|                                     | 675   | Kg        | 1122                                         |

| Parameter          | Value | Units | Notes                                                                   |
|--------------------|-------|-------|-------------------------------------------------------------------------|
| Mass (dry)         | 495   | Kg    | 1122C                                                                   |
|                    | 483   | Kg    | 1122K                                                                   |
|                    | 475   | Kg    | 1122                                                                    |
| Omni Antenna range | 1875  | Km    | assumes root model, 0.5 range multiplier and 0.5 multiple antenna bonus |

## Power budget

All consumption in EC/second.

| System           | Unit consumption | Amount | Total consumption | Notes |
|------------------|------------------|--------|-------------------|-------|
| Communotron 16   | 0.13             | 2      | 0.26              |       |
| HG-5             | 0.55             | 2      | 1.10              |       |
| Mission subtotal |                  |        | 1.36              |       |
| OKTO             | 0.02             | 1      | 0.02              |       |
| Systems subtotal |                  |        | 1.38              |       |
| Reaction wheel   | 0.03             | 1      | 0.03              |       |
| Abs. maximum     |                  |        | 1.41              |       |

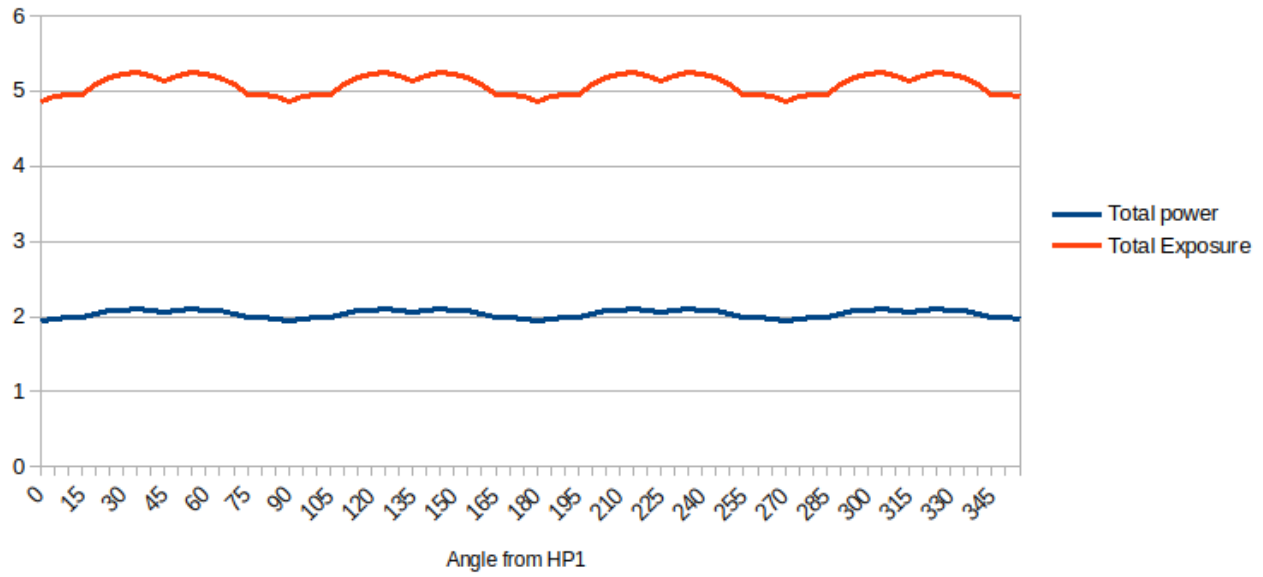
## Electrical generator performance

All figures assume side of power rings pointing towards the sun. Derate by cosine of angle to perpendicular of panels in other cases. Mission attitude is satellite perpendicular to ecliptic (direction measured as a line intersecting centre of computer section and centre of propulsion section).

Zero angle is at hardpoint 1 (where the computer label is).

# Model 1122 communications satellite

## Power generation



| Angle | Total power | Total Exposure | UPSR | T2R  | LPSR |
|-------|-------------|----------------|------|------|------|
| 0     | 1.94        | 4.86           | 1.93 | 1    | 1.93 |
| 5     | 1.97        | 4.93           | 1.88 | 1.08 | 1.97 |
| 10    | 1.98        | 4.96           | 1.81 | 1.16 | 1.99 |
| 15    | 1.98        | 4.95           | 1.73 | 1.22 | 2    |
| 20    | 2.03        | 5.08           | 1.81 | 1.28 | 1.99 |
| 25    | 2.07        | 5.18           | 1.88 | 1.33 | 1.97 |
| 30    | 2.09        | 5.23           | 1.93 | 1.37 | 1.93 |
| 35    | 2.1         | 5.24           | 1.97 | 1.39 | 1.88 |
| 40    | 2.08        | 5.21           | 1.99 | 1.41 | 1.81 |
| 45    | 2.06        | 5.14           | 2    | 1.41 | 1.73 |
| 50    | 2.08        | 5.21           | 1.99 | 1.41 | 1.81 |
| 55    | 2.1         | 5.24           | 1.97 | 1.39 | 1.88 |
| 60    | 2.09        | 5.23           | 1.93 | 1.37 | 1.93 |
| 65    | 2.07        | 5.18           | 1.88 | 1.33 | 1.97 |
| 70    | 2.03        | 5.08           | 1.81 | 1.28 | 1.99 |
| 75    | 1.98        | 4.95           | 1.73 | 1.22 | 2    |
| 80    | 1.98        | 4.96           | 1.81 | 1.16 | 1.99 |
| 85    | 1.97        | 4.93           | 1.88 | 1.08 | 1.97 |
| 90    | 1.94        | 4.86           | 1.93 | 1    | 1.93 |
| 95    | 1.97        | 4.93           | 1.97 | 1.08 | 1.88 |
| 100   | 1.98        | 4.96           | 1.99 | 1.16 | 1.81 |
| 105   | 1.98        | 4.95           | 2    | 1.22 | 1.73 |
| 110   | 2.03        | 5.08           | 1.99 | 1.28 | 1.81 |
| 115   | 2.07        | 5.18           | 1.97 | 1.33 | 1.88 |
| 120   | 2.09        | 5.23           | 1.93 | 1.37 | 1.93 |
| 125   | 2.1         | 5.24           | 1.88 | 1.39 | 1.97 |
| 130   | 2.08        | 5.21           | 1.81 | 1.41 | 1.99 |
| 135   | 2.06        | 5.14           | 1.73 | 1.41 | 2    |

|     |      |      |      |      |      |
|-----|------|------|------|------|------|
| 140 | 2.08 | 5.21 | 1.81 | 1.41 | 1.99 |
| 145 | 2.1  | 5.24 | 1.88 | 1.39 | 1.97 |
| 150 | 2.09 | 5.23 | 1.93 | 1.37 | 1.93 |
| 155 | 2.07 | 5.18 | 1.97 | 1.33 | 1.88 |
| 160 | 2.03 | 5.08 | 1.99 | 1.28 | 1.81 |
| 165 | 1.98 | 4.95 | 2    | 1.22 | 1.73 |
| 170 | 1.98 | 4.96 | 1.99 | 1.16 | 1.81 |
| 175 | 1.97 | 4.93 | 1.97 | 1.08 | 1.88 |
| 180 | 1.94 | 4.86 | 1.93 | 1    | 1.93 |
| 185 | 1.97 | 4.93 | 1.88 | 1.08 | 1.97 |
| 190 | 1.98 | 4.96 | 1.81 | 1.16 | 1.99 |
| 195 | 1.98 | 4.95 | 1.73 | 1.22 | 2    |
| 200 | 2.03 | 5.08 | 1.81 | 1.28 | 1.99 |
| 205 | 2.07 | 5.18 | 1.88 | 1.33 | 1.97 |
| 210 | 2.09 | 5.23 | 1.93 | 1.37 | 1.93 |
| 215 | 2.1  | 5.24 | 1.97 | 1.39 | 1.88 |
| 220 | 2.08 | 5.21 | 1.99 | 1.41 | 1.81 |
| 225 | 2.06 | 5.14 | 2    | 1.41 | 1.73 |
| 230 | 2.08 | 5.21 | 1.99 | 1.41 | 1.81 |
| 235 | 2.1  | 5.24 | 1.97 | 1.39 | 1.88 |
| 240 | 2.09 | 5.23 | 1.93 | 1.37 | 1.93 |
| 245 | 2.07 | 5.18 | 1.88 | 1.33 | 1.97 |
| 250 | 2.03 | 5.08 | 1.81 | 1.28 | 1.99 |
| 255 | 1.98 | 4.95 | 1.73 | 1.22 | 2    |
| 260 | 1.98 | 4.96 | 1.81 | 1.16 | 1.99 |
| 265 | 1.97 | 4.93 | 1.88 | 1.08 | 1.97 |
| 270 | 1.94 | 4.86 | 1.93 | 1    | 1.93 |
| 275 | 1.97 | 4.93 | 1.97 | 1.08 | 1.88 |
| 280 | 1.98 | 4.96 | 1.99 | 1.16 | 1.81 |
| 285 | 1.98 | 4.95 | 2    | 1.22 | 1.73 |
| 290 | 2.03 | 5.08 | 1.99 | 1.28 | 1.81 |
| 295 | 2.07 | 5.18 | 1.97 | 1.33 | 1.88 |
| 300 | 2.09 | 5.23 | 1.93 | 1.37 | 1.93 |
| 305 | 2.1  | 5.24 | 1.88 | 1.39 | 1.97 |
| 310 | 2.08 | 5.21 | 1.81 | 1.41 | 1.99 |
| 315 | 2.06 | 5.14 | 1.73 | 1.41 | 2    |
| 320 | 2.08 | 5.21 | 1.81 | 1.41 | 1.99 |
| 325 | 2.1  | 5.24 | 1.88 | 1.39 | 1.97 |
| 330 | 2.09 | 5.23 | 1.93 | 1.37 | 1.93 |
| 335 | 2.07 | 5.18 | 1.97 | 1.33 | 1.88 |
| 340 | 2.03 | 5.08 | 1.99 | 1.28 | 1.81 |
| 345 | 1.98 | 4.95 | 2    | 1.22 | 1.73 |
| 350 | 1.98 | 4.96 | 1.99 | 1.16 | 1.81 |
| 355 | 1.97 | 4.93 | 1.97 | 1.08 | 1.88 |

## Assembly instructions

The satellite is based on the SpaceBus 1 satellite design and can be built using the standard parts. The following table presents the modules stacked in order, with angle of rotation relative to the computer module.

| Module   | Type     | Angle | Notes                     |
|----------|----------|-------|---------------------------|
| Computer | Computer | 0     | Module defines zero angle |

| Module     | Type              | Angle | Notes                       |
|------------|-------------------|-------|-----------------------------|
| UPS        | Type 1 PSU        | 15    | HG-5 rest on top of OX-STAT |
| T2         | Type 2 PSU        | 0     |                             |
| LPS        | Type 1 PSU        | 45    |                             |
| Propulsion | Type 1 propulsion | 0     |                             |

| Hardpoint | Device       | Notes                           |
|-----------|--------------|---------------------------------|
| 1         | HG-5 antenna | Use 2x symmetry. Apply on HP 1. |
| 2         | HG-5 antenna | Use 2x symmetry. Apply on HP 1. |
| 3         | Empty        |                                 |
| 4         | Empty        |                                 |
| 5         | Empty        |                                 |
| 6         | Empty        |                                 |

Apply 8x Z-100 batteries on propulsion unit hardpoints, light facing towards exhaust.